



TRINITY GRAMMAR SCHOOL

Mathematics Department

2012

HALF YEARLY
EXAMINATION

HSC ASSESSMENT TASK 3

Year 12

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11 – 14
- Write your Board of Studies Student Number **and** Class Teacher on the writing booklet(s) **or** sheet(s) submitted
- **Weighting:** 30%

Total marks – 70

Section I

Pages 3 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II

Pages 7 – 14

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hour 45 minutes for this section

Section I 10 marks

- Circle the correct response on the answer sheet provided
 - Each question is worth 1 mark
-

1 The exact value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is:

(A) $\frac{\pi}{3}$

(B) $-\frac{\pi}{3}$

(C) $\frac{2\pi}{3}$

(D) $-\frac{2\pi}{3}$

2 State the domain for which the function $y = 4x - x^2$ is an increasing, one-to-one function.

(A) $x \leq 2$

(B) $x < 2$

(C) $x > 0$

(D) $x < 4$

3 What is the value of $\lim_{x \rightarrow 0} \frac{\sin 3x}{x}$?

(A) $\frac{1}{3}$

(B) 1

(C) 3

(D) undefined

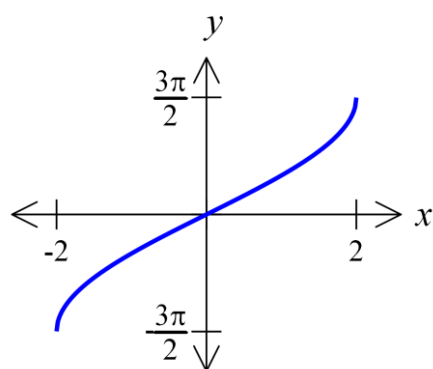
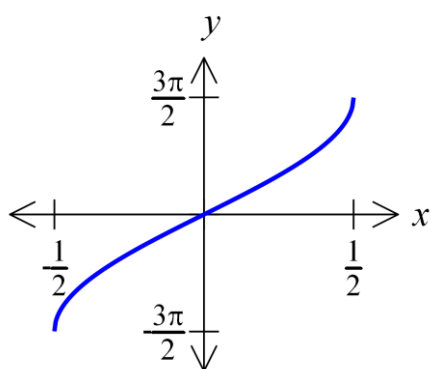
- 4 The domain of the function $y = \ln(x^2 + e)$ is:
- (A) $x \geq 0$
- (B) $x \leq 0$
- (C) all real values of x
- (D) all real $x, x \neq 0$
- 5 The range of the function $y = \ln(x^2 + e)$ is:
- (A) $y \geq 0$
- (B) $y \geq 1$
- (C) $y \geq e$
- (D) all real $y, y \neq 0$
- 6 The area under the curve $y = \sin x$ between $x = -\frac{\pi}{2}$, $x = \frac{\pi}{2}$ and the x - axis is:
- (A) -1
- (B) 0
- (C) 1
- (D) 2
- 7 To find $\int_0^1 x\sqrt{x^2 + 1} \, dx$ using the substitution $u = x^2 + 1$, in terms of u the correct expression is:
- (A) $\int_0^1 \sqrt{u} \, du$
- (B) $\frac{1}{2} \int_0^1 \sqrt{u} \, du$
- (C) $\int_1^2 \sqrt{u} \, du$
- (D) $\frac{1}{2} \int_1^2 \sqrt{u} \, du$

8 The exact value of $\sin\left(\cos^{-1}\left(\frac{4}{5}\right) + \tan^{-1}\left(-\frac{4}{3}\right)\right)$ is:

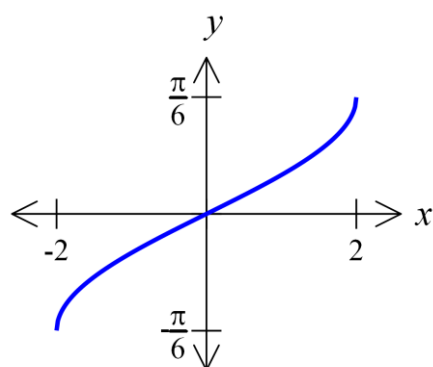
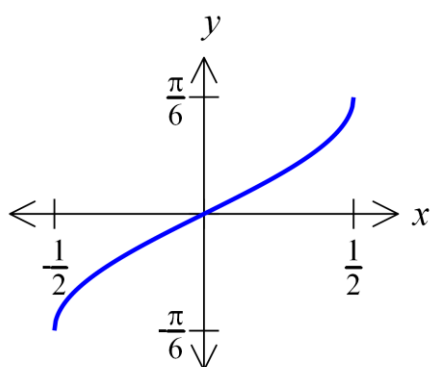
- (A) -1
- (B) $-\frac{7}{25}$
- (C) $\frac{7}{25}$
- (D) 1

9 Which of the graphs below represents $y = 3\sin^{-1}\frac{x}{2}$?

- (A) (C)



- (B) (D)



- 10 Below is the solution to the integral $\int_0^{\frac{\pi}{4}} \cos^2 x \, dx$.

$$\begin{aligned} & \int_0^{\frac{\pi}{4}} \cos^2 x \, dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{4}} (1 + \cos 2x) \, dx && \text{LINE I} \\ &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}} && \text{LINE II} \\ &= \frac{1}{2} \left[\left(\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right) - \left(0 - \frac{1}{2} \sin 0 \right) \right] && \text{LINE III} \\ &= \frac{\pi}{8} - \frac{1}{4} && \text{LINE IV} \end{aligned}$$

The line containing the FIRST mistake is:

- (A) LINE I
- (B) LINE II
- (C) LINE III
- (D) LINE IV

End of Section I

Section II 60 marks

- Begin each question in a new writing booklet or on a new answer sheet
 - Show all necessary working
 - Each question is worth 15 marks
-

Question 11 (15 marks)

a) Find:

i) $\frac{d}{dx} \sin(3x - 2)$ **1**

ii) $\int \sec^2 3x \, dx$ **1**

iii) $\int \sin^2 2x \, dx$. **2**

b) Find the value of $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$ **2**

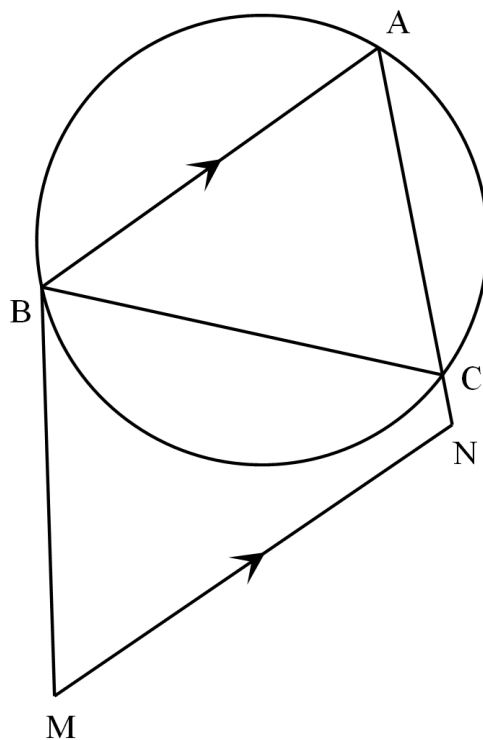
c)

i) Show that $\frac{1 + \cos 2x}{\sin 2x} = \cot x$. **2**

ii) Hence, find the exact value of $\cot 15^\circ$. **1**

d) Use the principle of mathematical induction to show that $9^{n+2} - 4^n$ is divisible by 5 for all positive integers of n . **3**

- e) ABC is a triangle inscribed in a circle. M is a point on the tangent to the circle at B and N is a point on AC produced so that MN is parallel to BA .



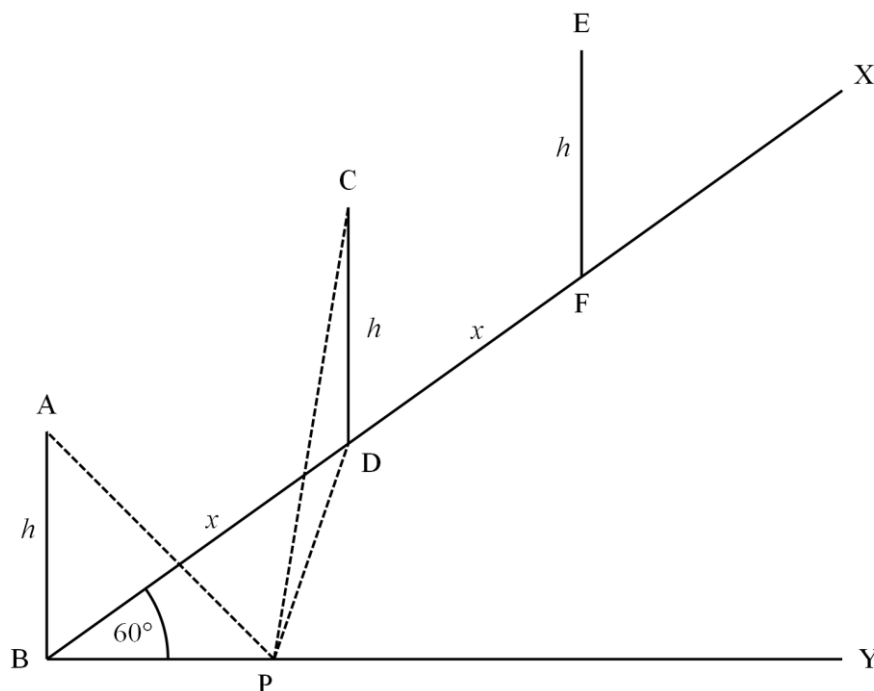
- i) Copy or trace the diagram into your answer booklet.
- ii) Give a reason why $\angle MBC = \angle BAC$.
- iii) Show that $MNCB$ is a cyclic quadrilateral.

1

2

Question 12 (15 marks) Begin a NEW answer booklet or answer sheet

- a) In the diagram below, BX and BY represent two roads intersecting at an angle of 60° . On the road BX are situated three telegraph poles AB , CD and EF , all of equal height, the same distance, x metres, apart (i.e. $BD = DF = x$). P is a point on the road BY and the angles of elevation to A and C are 45° and 30° respectively.



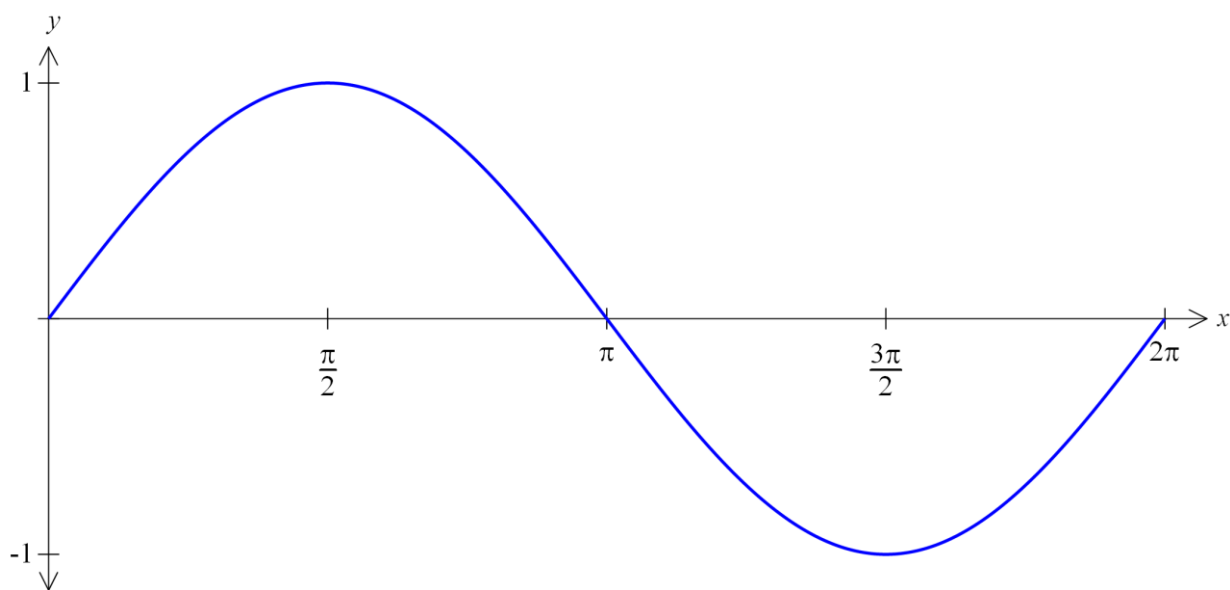
- i) Show that $DP = h\sqrt{3}$. 1
 - ii) Show that $\angle BDP = 30^\circ$ and hence that $\triangle BPD$ is right angled at P . 2
 - iii) Prove that $x = 2h$. 1
 - iv) Show that $PF = h\sqrt{13}$ and hence show that the angle of elevation of E from P is approximately 15.5° . 3
- b) State the domain and range of $y = \sin^{-1}(2x + 1)$ and then sketch the curve. 2

- c) The position x cm of a particle moving along an x -axis is given by $x = 3t + e^{-2t}$ where t is the time in seconds.
- i) What is the position of the particle when $t = \frac{1}{2}$ second? Give your answer in exact form. **1**
 - ii) What is the initial velocity of the particle? **2**
 - iii) Show the initial acceleration of the particle is 4 cm/s^2 . **2**
 - iv) Explain why the particle will never come to rest. **1**

Question 13 (15 marks) **Begin a NEW answer booklet or answer sheet**

- a) Find the slope of the tangent to $y = \tan^{-1}(\sqrt{1-x})$ at the point $y = \frac{\pi}{4}$. **2**
- b) Solve $\frac{2}{2-x} \geq 3$. **2**
- c) Evaluate $\int_0^{\frac{\pi}{6}} \frac{2\cos x}{1+4\sin^2 x} dx$ using the substitution $u = 2\sin x$. **4**
- d) The region bounded by the curve $y = \sin x$, the x -axis and the lines $x = \frac{\pi}{12}$ and $x = \frac{\pi}{4}$ is rotated through one complete revolution about the x -axis. Find the volume of the solid formed. **3**
- e) The curves $y = \ln x$ and $xy = e$ intersect at the point P ($e, 1$). Show that at the point P $\tan \theta = \frac{2e}{e^2 - 1}$. **2**

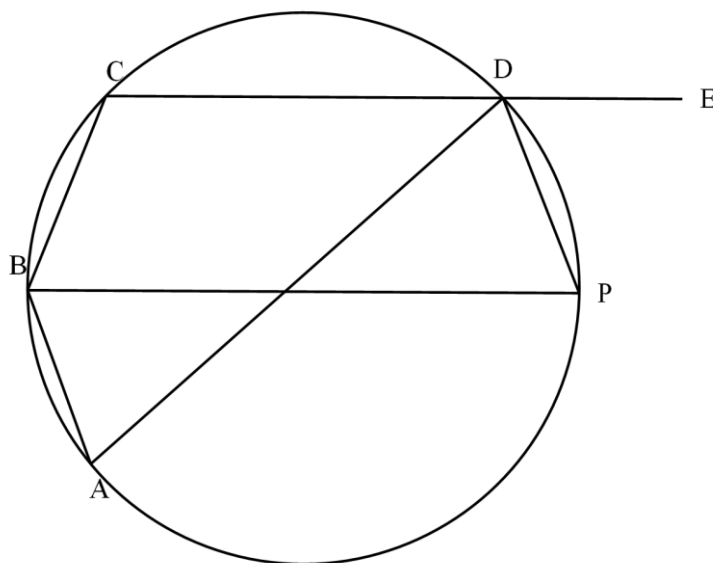
- f) The diagram shows the graph of $y = \sin x$ for $0 \leq x \leq 2\pi$. Copy or trace the diagram onto your answer sheet.



- i) On the same set of axes, sketch the graph of $y = \cos 2x$ for $0 \leq x \leq 2\pi$. **1**
- ii) Hence, state the number of solutions in $0 \leq x \leq 2\pi$ to the equation $\sin x = \cos 2x$. **1**

Question 14 **(15 marks)** **Begin a NEW answer booklet or answer sheet**

- a) An isotope of carbon, C_{14} decays at a rate proportional to the mass present. The rate of change is given by $\frac{dM}{dt} = -kM$ where k is a positive constant and M is the mass present.
- i) Show $M = M_0 e^{-kt}$ is a solution to this equation. **1**
- ii) The half-life of this isotope is 5600 years. This means it takes 5600 years for 100 grams of C_{14} to decay to 50 grams. Find the value of k correct to 3 significant figures. **1**
- iii) Archaeologists use radiocarbon dating to establish the age of discoveries. Calculate the age of an item in which only one-eighth of the original carbon remains. **1**
- b) In the diagram below $ABCD$ is a cyclic quadrilateral. CD is produced to E . P is a point on the circle through A, B, C, D such that $\angle ABP = \angle PBC$.



Copy or trace the diagram into your answer booklet marking all the information given.

- i) Explain why $\angle ABP = \angle ADP$. **1**
- ii) Show that PD bisects $\angle ADE$. **2**
- iii) If, in addition, $\angle BAP = 90^\circ$ and $\angle APD = 90^\circ$, explain where the centre of the circle is located. **1**

- c) A is the point $(-2, 1)$ and B is the point (x, y) . The point $P (13, -9)$ divides AB externally in the ratio $5:3$. Find the values of x and y . 2
- d) Find the area under the curve $y = \cos^{-1} x$ between $x=0$, $x = \frac{1}{2}$ and the $x -$ axis. 2
- e) For the function $f(x) = \frac{1}{1-x^2}$
- i) Find the inverse function. 2
- ii) Using Simpsons Rule, find the area under the inverse function between $x = 1$ and $x = 3$ using five function values. 2

End of Section II

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(\sqrt{x^2 - a^2} + x \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(\sqrt{x^2 + a^2} + x \right)$$

Note $\ln x = \log_e x, \quad x > 0$



TRINITY GRAMMAR SCHOOL
2012, Year 12 Mathematics Extension 1
Half Yearly Examination
SECTION I
ANSWER SHEET

Student Number: **Class Teacher:**

Be sure to write your answers for Section I on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. Mark only one answer for each question.

-
- | | | | | |
|-------------|---|---|---|---|
| Q1. | A | B | C | D |
| Q2. | A | B | C | D |
| Q3. | A | B | C | D |
| Q4. | A | B | C | D |
| Q5. | A | B | C | D |
| Q6. | A | B | C | D |
| Q7. | A | B | C | D |
| Q8. | A | B | C | D |
| Q9. | A | B | C | D |
| Q10. | A | B | C | D |



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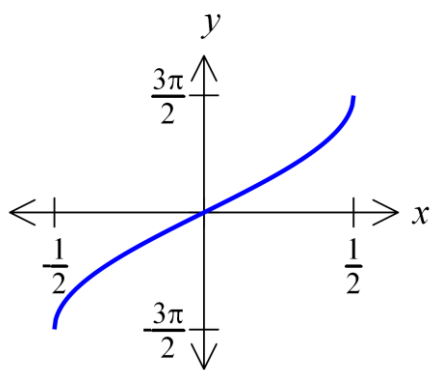
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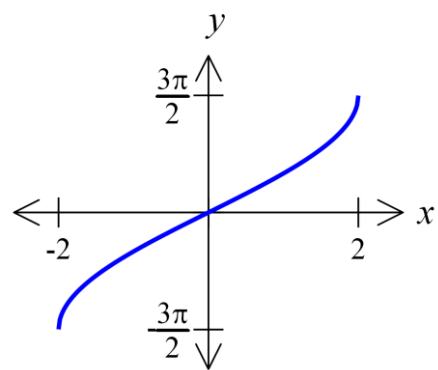
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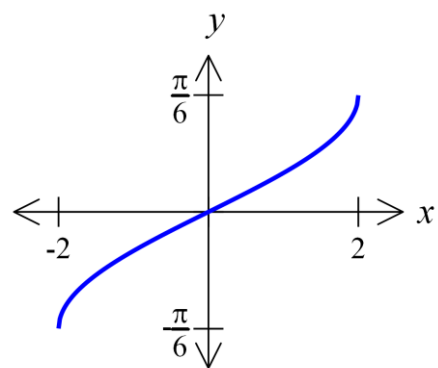
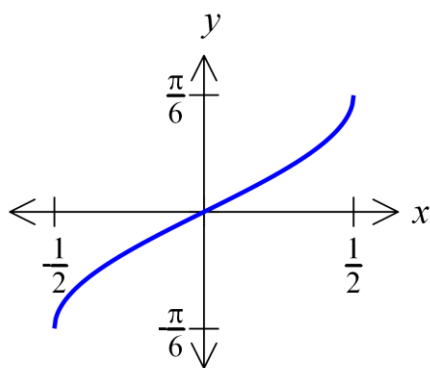
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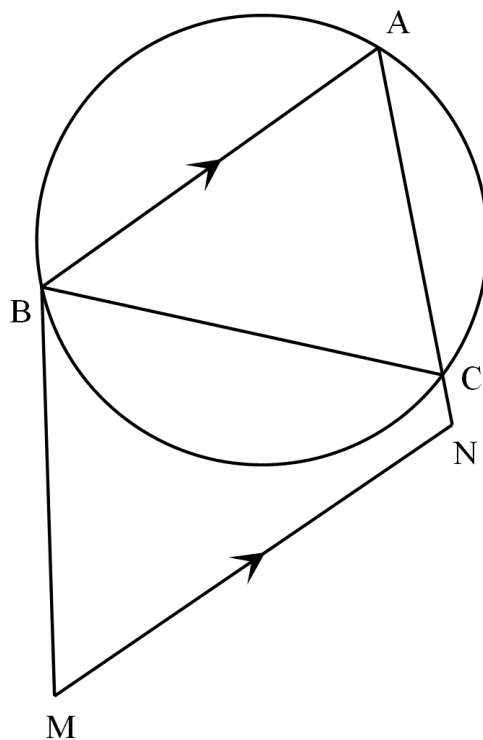
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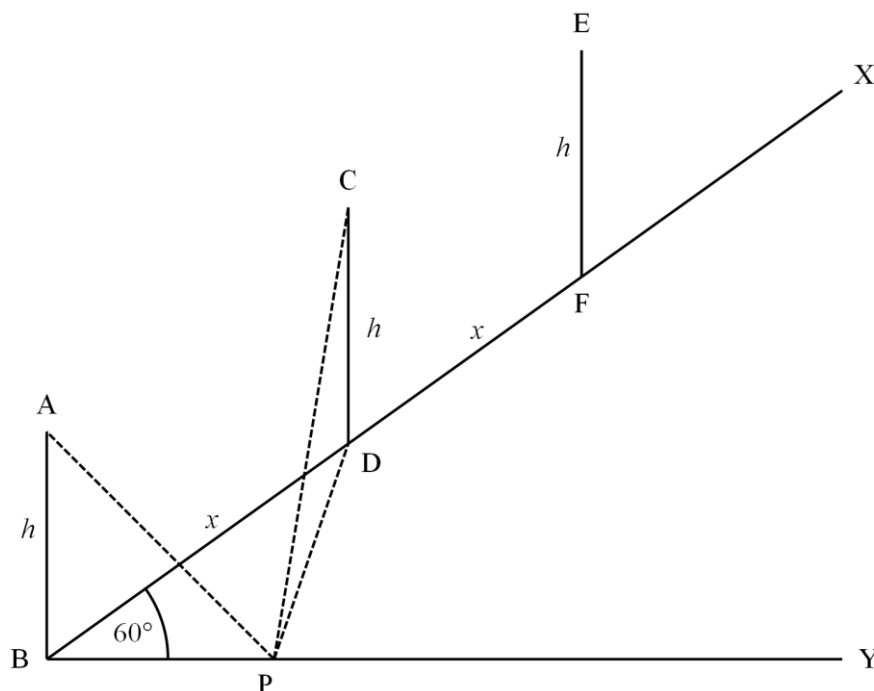
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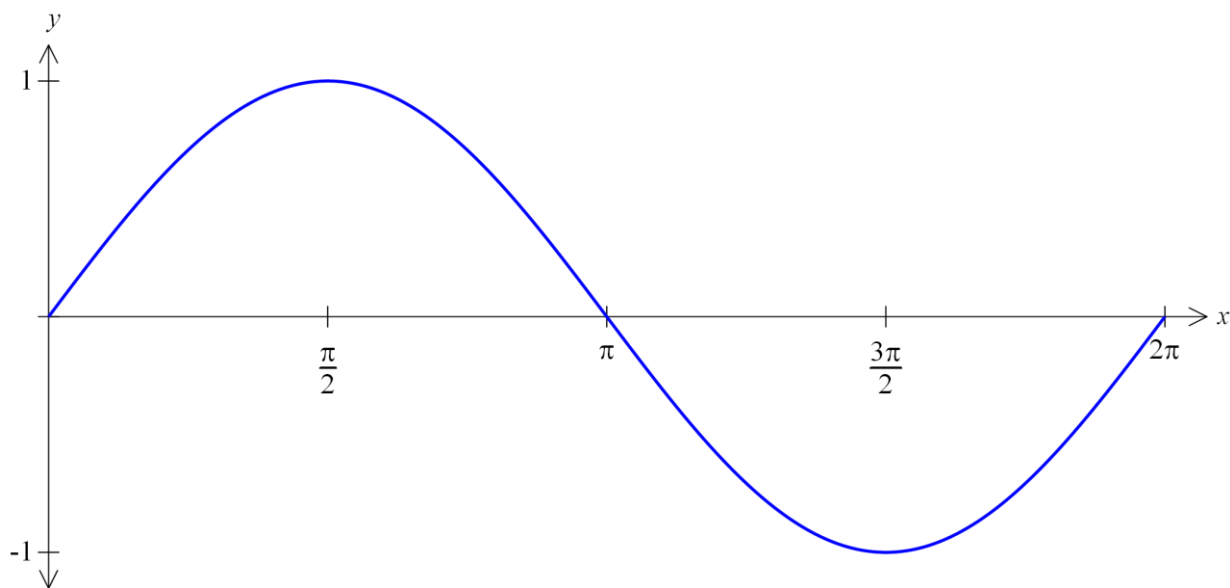
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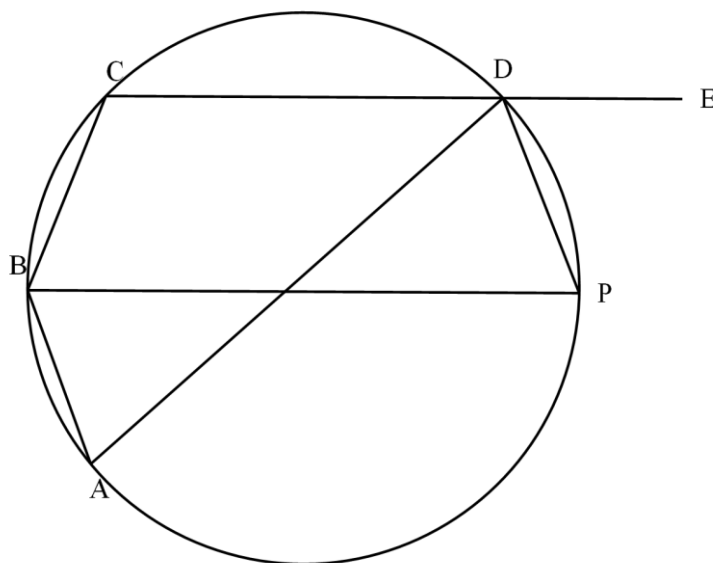
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- i) On the same set of axes, sketch the graph of $y = \cos 2x$ for $0 \leq x \leq 2\pi$. **1**
- ii) Hence, state the number of solutions in $0 \leq x \leq 2\pi$ to the equation $\sin x = \cos 2x$. **1**

Question 14 **(15 marks)** **Begin a NEW answer booklet or answer sheet**

- a) An isotope of carbon, C_{14} decays at a rate proportional to the mass present. The rate of change is given by $\frac{dM}{dt} = -kM$ where k is a positive constant and M is the mass present.
- i) Show $M = M_0 e^{-kt}$ is a solution to this equation. **1**
- ii) The half-life of this isotope is 5600 years. This means it takes 5600 years for 100 grams of C_{14} to decay to 50 grams. Find the value of k correct to 3 significant figures. **1**
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- e) For the function $f(x) = \frac{1}{1-x^2}$
- i) Find the inverse function. 2
- ii) Using Simpsons Rule, find the area under the inverse function between $x = 1$ and $x = 3$ using five function values. 2

End of Section II

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(\sqrt{x^2 - a^2} + x \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(\sqrt{x^2 + a^2} + x \right)$$

Note $\ln x = \log_e x, \quad x > 0$



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SECTION I
ANSWER SHEET

Student Number: **Class Teacher:**

Be sure to write your answers for Section I on this answer sheet. After you have selected an answer, **CIRCLE** the correct answer. To change an answer, erase your previous mark completely, and then record your new answer. Mark only one answer for each question.

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- | | | | | |
|-------------|---|---|---|---|
| Q1. | A | B | C | D |
| Q2. | A | B | C | D |
| Q3. | A | B | C | D |
| Q4. | A | B | C | D |
| Q5. | A | B | C | D |
| Q6. | A | B | C | D |
| Q7. | A | B | C | D |
| Q8. | A | B | C | D |
| Q9. | A | B | C | D |
| Q10. | A | B | C | D |



TRINITY GRAMMAR SCHOOL
2012, Year 12 Mathematics Extension 1
Half Yearly Examination
SECTION I
ANSWER SHEET

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Question 11

a) i) $\frac{d}{dz} = 3 \cos(3z-2) \checkmark$

ii) $\int \sec^2 3z dz$ Do not
 $= \frac{1}{3} \tan 3z + C \checkmark$ penalise for
 C

iii) $\int \sin^2 2x dx$
 $= \frac{1}{2} \int 1 - \cos 4x dx \checkmark$
 $= \frac{1}{2} \left[x - \frac{1}{4} \sin 4x + C \right] \checkmark$
 $= \frac{1}{2} x - \frac{1}{8} \sin 4x + C$

b) $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$
 $= \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} \checkmark$
 $= 2 \cdot 1 \cdot 1 = 2 \checkmark$

c) i) LHS = $\frac{1 + \cos 2x}{\sin 2x}$
 $= \frac{2 \cos^2 x}{2 \sin x \cos x} \checkmark$
 $= \frac{\cos x}{\sin x}$
 $= \cot x = \text{RHS}$

ii) $\cot 15^\circ = \frac{1 + \cos 2(15)}{\sin 2(15)}$
 $= \frac{1 + \cos 30}{\sin 30}$
 $= \left(1 + \frac{\sqrt{3}}{2}\right) \div \frac{1}{2} = 2 + \sqrt{3} \checkmark$

el) For $n=1$
 $9^{n+2} - 4^n$
 $= 9^{1+2} - 4^1$
 $= 9^3 - 4$
 $= 725$

which is divisible by 5. $\checkmark$

Assume that when $n=k$,
 $9^{k+2} - 4^k$ is divisible by 5.

i.e. $9^{k+2} - 4^k = 5M \checkmark$

where M is an integer.

or $9^{k+2} = 5M + 4^k$

When $n = k+1$,

$9^{k+1+2} - 4^{k+1}$
 $= 9 \cdot 9^{k+2} - 4 \cdot 4^k$
 $= 9(5M + 4^k) - 4 \cdot 4^k$
 $= 9 \cdot 5M + 9 \cdot 4^k - 4 \cdot 4^k$
 $= 5 \cdot 9M + 5 \cdot 4^k$
 $= 5(9M + 4^k) \checkmark$

Since $9M + 4^k$ is an integer.

$5(9M + 4^k)$ is divisible by 5.

$\therefore$ By the process of M.I.
 it has been shown that
 $9^{n+2} - 4^n$ is divisible by 5 for all $n \geq 1$

Question 11 cont.

e) ii) the angle between a tangent & a chord is equal to the angle subtended by the chord in the alternate segment. ✓

$$\text{iii) } \begin{aligned} \hat{MNC} &= 180 - \hat{BAC} \quad (\text{co-int } \angle\text{'s, } AB \parallel NM) \\ \therefore \hat{MNC} &= 180 - \hat{MBC} \quad (\text{from above, } \hat{BAC} = \hat{MBC}) \end{aligned} \quad \checkmark$$

$\therefore$ MBCN is cyclic quad, opposite angles supplementary. ✓

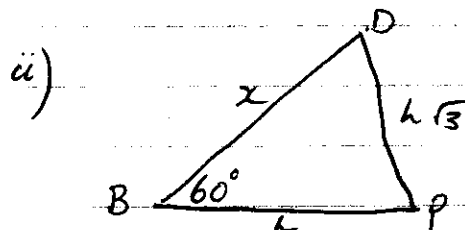
Question 12.

a) i) In $\triangle CDP$, $\hat{CPD} = 30^\circ$

$$\therefore \tan 30 = \frac{h}{DP}$$

$$\frac{1}{\sqrt{3}} = \frac{h}{DP}$$

$$DP = \sqrt{3}h.$$



BP in right angle $\triangle$.

$$\frac{\sin \hat{BDP}}{L} = \frac{\sin 60}{h\sqrt{3}}$$

$$\sin \hat{BDP} = \frac{K \cdot \frac{\sqrt{3}}{2}}{\frac{1}{K\sqrt{3}}}$$

$$\sin \hat{BDP} = \frac{1}{2}$$

$$\therefore \hat{BDP} = 30^\circ$$

$$\therefore \hat{BPD} = 90^\circ \quad (\angle \text{sum } \triangle)$$

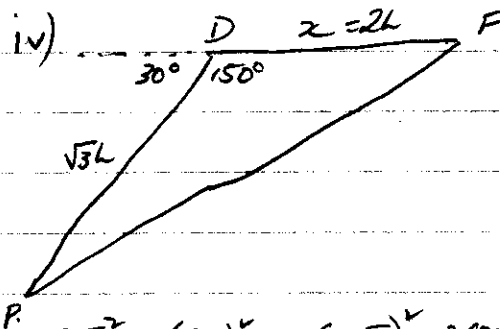
iii) In $\triangle BPD$

$$x^2 = (h\sqrt{3})^2 + L^2$$

$$= 3h^2 + L^2$$

$$2L^2 = 4h^2$$

$$\therefore x = 2h.$$

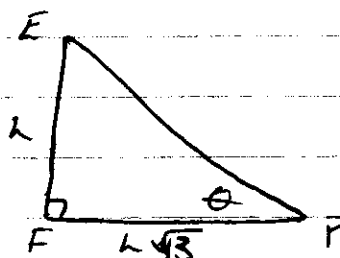


$$PF^2 = (2h)^2 + (h\sqrt{3})^2 - 2(2h)(h\sqrt{3})\cos 150$$

$$= 7h^2 - 2\sqrt{3}h^2 \left(-\frac{\sqrt{3}}{2}\right)$$

$$= 13h^2$$

$$PF = h\sqrt{13}$$



$$\tan \theta = \frac{L}{L\sqrt{3}}$$

$$\tan \theta = 0.277$$

$$\theta = 15.5^\circ$$

b) $y = \sin^{-1}(2x+1)$

Domain:

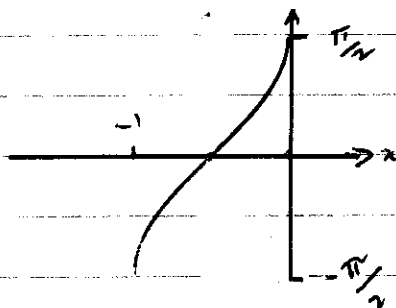
$$-1 \leq 2x+1 \leq 1$$

$$-2 \leq 2x \leq 0$$

$$-1 \leq x \leq 0$$

Range

$$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$



Q12

c)

$$x = 3t + e^{-2t}$$

$$\dot{x} = 3 - 2e^{-2t}$$

$$\ddot{x} = 4e^{-2t}$$

i) at $t = \frac{1}{2}$

$$x = 3\left(\frac{1}{2}\right) + e^{-2\left(\frac{1}{2}\right)}$$

$$= \frac{3}{2} + e^{-1} \text{ or } \frac{3}{2} + \frac{1}{e}$$

ii) initial velocity at $t=0$

$$\dot{x} = 3 - 2e^{-2(0)} \quad \checkmark$$

$$= 3 - 2$$

$$= 1 \text{ cm/s. } \checkmark$$

iii) initial accel. at $t=0$

$$\ddot{x} = 4e^{-2(0)} \quad \checkmark$$

$$= 4 \text{ cm/s}^2 \quad \checkmark$$

iv) the acceleration $4e^{-2t}$ is

always greater than zero, $\therefore$
so the particle is always
speeding up. $\checkmark$

Question 13

a) $y = \tan^{-1}(\sqrt{1-x})$

$$\therefore \sqrt{1-x} = \tan y$$

$$1-x = \tan^2 y$$

$$x = 1 - \tan^2 y$$

$$\frac{dx}{dy} = -2 \sec^2 y \tan y \quad \checkmark$$

at $y = \frac{\pi}{4}$

$$\frac{dx}{dy} = -2 \sec^2\left(\frac{\pi}{4}\right) \tan\left(\frac{\pi}{4}\right)$$

$$= -2 \left(\frac{\sqrt{2}}{1}\right)^2 \cdot 1$$

$$= -4$$

$$\therefore m = \frac{dy}{dx} = \frac{1}{\frac{dx}{dy}} = -\frac{1}{4} \quad \checkmark$$

b) $\frac{2}{2-x} \geq 3$

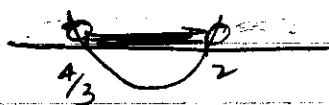
$$2(2-x) \geq 3(2-x)$$

$$4-2x \geq 12-12x+3x^2$$

$$0 \geq x^2 - 10x + 8$$

$$0 \geq 3x^2 - 6x - 4x + 8$$

$$0 \geq (3x-4)(x-2) \quad \checkmark$$



$$\therefore \frac{4}{3} \leq x \leq 2 \quad \checkmark$$

c) $\int_0^{\pi/6} \frac{2\cos x}{1+4\sin^2 x} dx$

$$u = 2\sin x$$

$$du = 2\cos x dx \quad \checkmark$$

at $x=0$ $u=0$

$x=\pi/6$ $u=1$ $\checkmark$

$$\therefore = \int_0^1 \frac{du}{1+u^2}$$

$$= [\tan^{-1} u]_0^1 \quad \checkmark$$

$$= \tan^{-1}(1) - \tan^{-1}(0)$$

$$= \frac{\pi}{4} - 0 = \frac{\pi}{4} \quad \checkmark$$

d) $V = \pi \int_{\pi/12}^{\pi/4} (\sin x)^2 dx$

$$= \frac{\pi}{2} \int_{\pi/12}^{\pi/4} 1 - \cos 2x dx \quad \checkmark$$

$$= \frac{\pi}{2} \left[x - \frac{1}{2} \sin 2x \right]_{\pi/12}^{\pi/4} \quad \checkmark$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{4} - \frac{1}{2} \sin 2 \cdot \frac{\pi}{4} \right) - \left(\frac{\pi}{12} - \frac{1}{2} \sin 2 \cdot \frac{\pi}{12} \right) \right]$$

$$= \frac{\pi}{2} \left(\frac{\pi}{4} - \frac{1}{2} - \frac{\pi}{12} + \frac{1}{4} \right)$$

$$= \frac{\pi^2}{12} - \frac{\pi}{8} \text{ units}^3 \quad \checkmark$$

Q13

e)

$$y = \ln x$$

$$\frac{dy}{dx} = \frac{1}{x}$$

at $x = e$.

$$m_1 = \frac{1}{e}$$

For

$$y = \frac{e}{x} = ex^{-1}$$

$$\frac{dy}{dx} = -ex^{-2}$$

at $x = e$.

$$m_2 = \frac{-e}{e^2} = -\frac{1}{e}$$

For angle between lines:

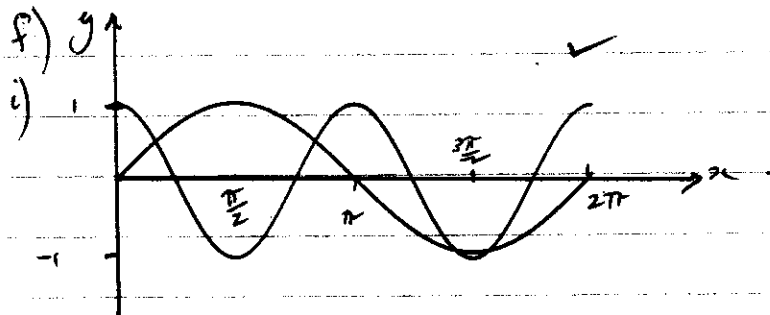
$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$= \left(\frac{1}{e} - \left(-\frac{1}{e} \right) \right) \div \left(1 + \frac{1}{e} \cdot -\frac{1}{e} \right)$$

$$= \frac{2}{e} \div \left(1 - \frac{1}{e^2} \right)$$

$$= \frac{2}{e} \times \frac{e^2}{e^2 - 1}$$

$$\therefore \tan \theta = \frac{2e}{e^2 - 1}$$



ii) 3 solutions ✓

Question 14

a) i) $\frac{dM}{dt} = -kM$

ii) $M = M_0 e^{-kt}$
 $\frac{dM}{dt} = -k M_0 e^{-kt}$
 $= -kM$ ✓

iii) $50 = 100 e^{-k(5600)}$
 $\frac{1}{2} = e^{-k(5600)}$
 $\ln \frac{1}{2} = -k(5600)$
 $-0.693 = -k(5600)$
 $k = \frac{0.693}{5600}$
 $= 0.000124$ ✓

iv) $\frac{M}{M_0} = e^{-kt}$
 $\frac{1}{8} = e^{-kt}$
 $\ln \frac{1}{8} = -kt$
 $-2.079 = -kt$
 $t = \frac{2.079}{k}$

$= 16800 \text{ years.}$ ✓

b) i) angles at the circumference standing on the same arc are equal. ✓

ii) $\angle PDE = \angle CBP$ (ext. $\angle$ of cyclic quadr equal to int. opp. $\angle$).

$\therefore \angle PDE = \angle ADP$

$\therefore PD$ bisects $\angle ADE$ ✓✓

iii) Since $\angle BAP = 90^\circ$, BP is a diameter ($\angle$ in a semi circle).
 Since $\angle AFD = 90^\circ$, AD is a diameter ($\angle$ in a semi circle).

$\therefore$ the centre is where BP & AD intersect. ✓

c) $13 = \frac{5x - 3(2)}{5 + (-3)}$
 $13 = \frac{5x - 6}{2}$

$26 = 5x - 6$

$5x = 32$

$5x = 32$

For y :

$-9 = \frac{5y - 3(1)}{2}$

$-18 = 5y - 3$

$5y = -15$

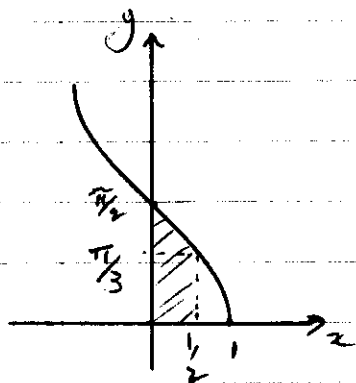
$5y = -15$ ✓

$\therefore B(4, -3)$

$(2, 1) : (x, y)$
 $5 : -3$

Q 14

d)



ii

x	1	1.5	2	2.5	3
y	0	0.577	$\frac{1}{\sqrt{2}}$	0.774	0.816

$$A = \frac{0.5}{3} \left(0 + 0.816 + 4 \left(0.577 + 0.774 \right) + 2 \left(\frac{1}{\sqrt{2}} \right) \right)$$

$$= 1.27 \text{ units}^2 \quad (2dp)$$

$$A = \int_0^{\pi/2} \cos^2 x \, dx$$

$$= \frac{1}{2} \times \frac{\pi}{2} + \int_{\pi/3}^{\pi/2} \cos y \, dy \quad \checkmark$$

$$= \frac{\pi}{6} + [\sin y]_{\pi/3}^{\pi/2}$$

$$= \frac{\pi}{6} + \left(\sin \frac{\pi}{2} - \sin \frac{\pi}{3} \right)$$

$$= \frac{\pi}{6} + \left(1 - \frac{\sqrt{3}}{2} \right)$$

$$= \frac{\pi}{6} + \frac{2 - \sqrt{3}}{2} \quad \checkmark$$

e) i $f(x): y = \frac{1}{1-x^2}$

$$f^{-1}(x): x = \frac{1}{1-y^2}$$

$$1-y^2 = \frac{1}{x}$$

$$y^2 = 1 - \frac{1}{x}$$

$$y = \sqrt{\frac{x-1}{x}}$$